Desk Watch JOY STUDENT

User guide in English

A compact desk console with a 16 x 2 character RGB LCD and a thumb joystick. It measures temperature and humidity and provides several practical tools.

The 10 menu entries

Entry	Function
Temp & Hum	Readings, min/max and average views, alert thresholds
Timer	Work sessions and breaks (Pomodoro)
Counter	Count, pause and reset
Stopwatch	Measure elapsed time
Alarm	Alarm and ringtone selection
Dice	Roll a die from 2 to 100 sides
Chess clock	Two player clocks with increment
Melody	Play four tunes
Settings	Adjust options and choose the startup favorite
Screen sleep	Turn off the display without disconnecting power

Startup favorite

The favorite is not a menu entry. Set it in Settings to choose the page shown at startup. Auto starts on the live Temp & Hum page.

Power on

Connect USB to a computer, charger or power bank. Do not touch the joystick during startup: its neutral position is calibrated then. Unplug USB to turn the console fully off.

Joystick gestures

Gesture	General action
Up / down	Browse choices or change a value. Hold to repeat.
Right or center click	Confirm / continue.
Left	Go back or cancel, depending on the page.
Double click	Page-specific action (see below).
Hold left for 2.5 s	Put the display to sleep.

Measurements and timer

Temp & Hum

The first line always shows the time (or --:-- if no RTC is detected). The second line shows temperature and humidity. Compatible sensors are Grove SHT31 and SHT40.

From the readings page	Action
Up / down	Cycle through live readings, min/max, then averages.
Right	Open the Tmin, Tmax, Hmin and Hmax threshold setup.
Left during setup	Go back one step, including from the confirmation page.
Right to advance each threshold	Adjust with up/down, then move to the next threshold.
Final “Select / OK ->” page	Click the joystick to save. The backlight confirms in green for 1.5 s.

The min/max view alternates between minimum and maximum values. The average view summarizes readings over time. When a reading crosses a threshold, an alert message appears and the backlight signals the condition.

Timer

At zero, use up/down to set the work duration: 1 to 60 minutes in 1-minute steps, then 65 minutes to 4 hours in 5-minute steps. The title simply reads TIMER; there is no session number.

Gesture	Action
Center click	Start, pause or resume.
Double click	Reset; if already at zero, return to the menu.
Open the menu	The countdown continues in the background.

Each work session is followed by a 5-minute short break. Every fourth session is followed by a 15-minute long break. The background color changes by phase and a sound marks each transition.

Counter, stopwatch and alarm

Counter

The first line reads COUNT ACTIVE or COUNT PAUSE. The second line shows the count, from 0 to 9999, between up and down arrows.

Gesture	Action
Click the joystick	Toggle between active and paused.
Up / down while active	Add / subtract 1.
Right	Open the reset confirmation.
Confirmation	Down = YES; up = NO.

Stopwatch

Center click starts the stopwatch, then pauses or resumes it. Double click resets it; double click while already at zero returns to the menu. Up and down do nothing. The stopwatch continues in the background if you open the menu.

Alarm

The RTC module is required. On entry, choose directly: up to enable the alarm, down to disable it. If disabled, a confirmation appears and the console returns to the menu after about 1.5 seconds. If enabled, set the hours, minutes, ringtone and volume in order. Use the joystick to go back a step.

After choosing the volume, “Alarm / OK” confirms the settings with a green flash for 1.5 s, then returns to the menu. There is no extra arming step. Eight ringtones are available: Birthday, Fur Elise, Mario Bros, Tetris, Aurora, Cosmic Pulse, Arcade Hero and Epic Rise.

When the alarm rings, a click or double click snoozes it for 5 minutes; a long press stops it. If there is no response, it stops after 2 minutes and is snoozed. The alarm can ring while the display is asleep.

Dice, chess clock and melody

Dice

Use up/down to choose the number of sides: 2, 4, 6, 8, 10, 12, 20, 30 or 100. Click to roll; the result stays on screen after the sound animation. Double click returns to the menu.

Chess clock

When stopped, choose a time control (starting minutes + increment seconds): 1+0, 3+0, 3+2, 5+0, 5+3, 10+0, 10+5, 15+10, 30+0 or 60+0. Click to start; Player 1 goes first.

During a game, move the joystick to switch turns. A beep confirms the change. Double click or a long press pauses the game. Click to resume; double click resets it. When a player runs out of time, the screen displays END and the buzzer sounds. Leaving the page pauses the game.

Melody

Use up/down to choose Birthday, Fur Elise, Mario Bros or Tetris. Click to play, pause or resume. Double click returns to the menu. Set the overall volume in Settings.

Colors and confirmation

The backlight helps identify each app and its state. A green flash confirms a saved setting. The exact shade may vary with the installed RGB display.

Tip

For multi-step settings, use left to go back and right or center click to continue. On a confirmation page, check the message before confirming.

Settings, sleep and troubleshooting

Settings

Use up/down to browse settings. Click to edit, then click again to save. Double click cancels the current edit and returns to the menu. Settings are kept when the console is unplugged.

Setting	Options
Brightness	8 levels, 5% to 100%, with live preview
Volume	Low, medium or high
Startup sound	On / off
Auto sleep	Never, 1, 2, 5, 10, 15 or 30 min, 1 h
Night mode	Off, automatic from 10 pm to 6 am (RTC), or always on
Set time	Set hours then minutes (RTC required)
Startup favorite	Auto, Temp & Hum, Clock, Timer, Stopwatch or Counter

Screen sleep

Choose Screen sleep in the menu or hold the joystick left for 2.5 seconds. Move or click the joystick to wake the display. Auto sleep follows the delay set in Settings and is suspended during a chess game. The alarm and background tasks can continue.

Quick troubleshooting

Symptom	Check
Blank display	Wake the console, then check USB, the LCD Grove cable and brightness.
Joystick points the wrong way	Restart without touching the joystick during calibration.
Sensor missing	Check the SHT31/SHT40 I2C connection, then restart.
Time “--:--” / RTC missing	Check the RTC module and I2C cable. Time-based features need the RTC.
No sound	Check the volume and the buzzer connected to D3.
Incorrect time	Open Settings > Set time (RTC required).
Repeated alert	Adjust Tmin, Tmax, Hmin and Hmax in Temp & Hum.